

PAPER **First-Principles Radiation Tolerance of Centrepost REBCO in a Compact Spherical-Tokamak Breeder: Chain-Oxygen-Vacancy Formation by DFT+U and YBCO Displacement-Cascade Molecular Dynamics**

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Abstract

The consumable centrepost of the Hyperion breeder carries the machine’s peak field of 16.84 T inside its harshest neutron environment, so the irradiation lifetime of the centrepost REBCO conductor—not stress and not quench—sets the cadence at which the plant is opened and, through it, the fraction of the calendar available for power production. We pose the lifetime question as a coupled multiscale problem and solve it from first principles. Steady 14 MeV neutron transport supplies the fast-flux field and the primary-knock-on-atom (PKA) spectrum at the conductor; a displacement-cascade molecular-dynamics (MD) treatment with a machine-learned interatomic potential and an arc-dpa surviving-defect model converts that spectrum into Frenkel-pair statistics resolved by sublattice; a DFT+U treatment of the correlated Cu 3d/O 2p manifold fixes the chain-oxygen-vacancy energetics; and a hole-doping dome maps chain-oxygen loss onto the critical temperature and current. A Monte-Carlo composition screen over 4,000 samples spanning the material and irradiation-model uncertainties finds that an *un-doped* YBCO tape reaches its critical-current break-even fluence at $\Phi^* = 5.64 \times 10^{22} \text{ n m}^{-2}$ —a factor of **1.48** above the artificial-pinning-center (APC) grade at $3.82 \times 10^{22} \text{ n m}^{-2}$ —and the ordering un-doped > APC is robust in **100** % of the samples. The displacement analysis localises ≈ 78 % of surviving lattice displacements on the oxygen sublattice, naming chain-oxygen disorder as the dominant lifetime-limiting channel; and a converged DFT+U calculation of the pristine cell ($E_{\text{tot}} = -2907.082 \text{ Ry}$) anchors the chain-oxygen-vacancy formation-energy series from a validated ground state. We report the governing equations, the numerical formulation, and the GPU high-performance-computing (HPC) campaign that produced the results, and we close with the one honest reconciliation the design must carry: the break-even-fluence ranking (radiation tolerance) is a distinct metric from the end-of-life in-field I_c margin at the full 16.84 T operating point, and the two are separated cleanly rather than conflated. The centrepost gains a physically grounded, conservative irradiation-lifetime basis whose engineering lever is oxygen-sublattice integrity, not pinning-center density.

Keywords: REBCO; YBCO; radiation damage; DFT+U; displacement cascade; oxygen vacancy; critical current; spherical-tokamak centrepost

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1 Introduction

The Hyperion breeder is a compact, high-field spherical tokamak (ST) whose central column is a demountable REBCO centrepost operating at a peak field of 16.84 T, behind a plasma of $I_p = 9.66$ MA at major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$ and negative triangularity $\delta = -0.30$, delivering $Q = 3.076$ at $P_{\text{fus}} = 85.04$ MW. The centrepost sits closest to the 14 MeV source and absorbs the highest neutron fluence of any magnet in the plant. Because the column is designed to be swapped rather than to last the plant lifetime, its *irradiation lifetime* is not a reliability footnote: it is the quantity that sets how often the machine is opened, and thus the availability envelope of the whole plant.

The governing question is quantitative and material. At what fast-neutron fluence does the conductor’s critical current fall to the design break-even point, and what physical process drives that decline? The two parts of the question are coupled: an actionable lifetime number is only as good as the mechanism behind it, because the mechanism names the design lever. This paper answers both, from electronic structure and atomistic simulation up to a lifetime figure of merit, and it does so for the specific conductor and field of the Hyperion centrepost.

1.1 Prior art

REBCO (rare-earth barium copper oxide; here $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$, “YBCO”) retains a high critical-current density in high field, which is why it is the enabling conductor for compact high-field magnets in the ARC and SPARC line of high-field tokamaks and for the ST pilot-plant class [2, 3, 1]. The coated-conductor manufacturing base has matured to the current densities and volumes a fusion magnet demands [4, 5], and record dc fields above 45 T have been demonstrated with no-insulation REBCO windings [6]. Fast-neutron irradiation degrades the in-field critical current of these conductors by introducing point defects and disorder that both suppress the superconducting order parameter and rearrange the flux-pinning landscape [10, 11, 12]. The engineering figure of merit is the *break-even fluence*: the accumulated fast-neutron fluence at which the irradiated critical current drops to the design floor.

Commercial REBCO is frequently supplied with *artificial pinning centers* (APC)—engineered nanoscale $\text{BaZrO}_3/\text{BaHfO}_3$ inclusions that raise the in-field critical current in the pristine state [8, 9, 7]. The design question is whether that pristine advantage survives irradiation, or whether the added disorder consumes the radiation-tolerance budget faster than it raises the starting current. We resolve this by direct comparison, and we anchor the resolution in the mechanism.

On the electronic-structure side, YBCO’s superconductivity is controlled by charge transfer from the Cu–O chains to the CuO_2 planes, so the oxygen content of the chains sets the hole doping and hence the critical temperature [13, 14, 15]. A correlated-electron treatment is required: the $\text{Cu } 3d/\text{O } 2p$ manifold is mishandled by a plain generalised-gradient functional, and the Hubbard-corrected DFT+U method is the standard first-principles remedy [17, 18, 19]. On the damage-production side, the connection from a neutron field to a displacement count runs through the Norgett–Robinson–Torrens (NRT) standard and its modern arc-dpa correction for athermal in-cascade recombination [24, 25], evaluated on cascades computed by large-scale molecular dynamics [23, 26] with interatomic forces from machine-learned potentials [27, 28, 29] trained on first-principles data.

1.2 Contribution

This paper is the first-principles complement to the centrepost magnet-integrity case. That case establishes that the column is structurally sound with a large margin and thermally stable, so the neutron budget of the conductor is the governing lifetime term rather than stress or quench [32, 33]; the present work supplies the physical content of that budget. Three results move together. (i) A 4,000-sample composition screen shows the un-doped grade is $1.48\times$ more radiation-tolerant than the APC grade on the break-even-fluence metric, and the ordering is universal. (ii) The damage is localised on the oxygen sublattice at $\approx 78\%$ of surviving displacements, so the lifetime-limiting mechanism is chain-oxygen disorder. (iii) The mechanism is anchored at the electronic-structure level by a converged DFT+U ground state, from which the chain-oxygen-vacancy energetics are referenced. We state each result as the output of an explicit governing model (§3) solved by a named GPU-HPC campaign (§4), and we place the numbers against the published irradiation literature (§6). The companion papers develop the conductor and winding basis [31], the un-doped-tape rule in the broader stabilizer context [34], the fast-fluence/dpa/shielding neutronics that set the flux [33], the fluence-driven remaining-useful-life scheduling [36], and the machine-learned-potential-to-transport materials pipeline whose methods this paper shares [37].

Two distinct magnets appear in the Kronos fleet and must not be conflated: the breeder centrepost analysed here, governed by its own 16.84 T peak field, and the burner plug coil at 26.49 T, engineered separately [39]. All results below refer to the centrepost conductor.

2 The centrepost lifetime problem

The centrepost is a demountable, radially thin REBCO winding forming the inboard leg of the toroidal-field circuit of a low-aspect-ratio ST. Its physical predicament is geometric: at $A = 2.5$ and $R_0 = 1.20$ m the inboard radial build is severely constrained, so the neutron shield between the plasma and the conductor is thin, and the fast flux at the winding is a small but non-negligible fraction of the first-wall value. The first wall itself sees $19.66 \text{ dpa fpy}^{-1}$ under 14 MeV loading [38]; the centrepost, behind its shield, accumulates a far lower displacement rate, but over a full-power year it is enough to be the binding lifetime constraint of the machine [33, 40].

The design response is to make the column consumable: a cartridge that is swapped on a defined cadence rather than qualified for the plant lifetime. That choice converts a materials-survival problem into a scheduling problem, and it makes the break-even fluence Φ^* the single most consequential conductor property, because the swap cadence is $\nu_{\text{swap}} = 1/\tau$ with lifetime $\tau = \Phi^*/\bar{\phi}$ at the mean centrepost fast flux $\bar{\phi}$. A larger Φ^* is a longer-lived centrepost, a lower cadence, and a higher plant availability. Everything in this paper is organised around computing Φ^* for two candidate grades and understanding the mechanism that sets it.

3 Governing equations and the formal problem

We formalise the lifetime problem as a chain of five coupled models, sketched in Figure 1: damage production, cascade partition, chain-oxygen accumulation, electronic structure, and the critical-current response. The break-even fluence is the functional of these that the composition screen samples.

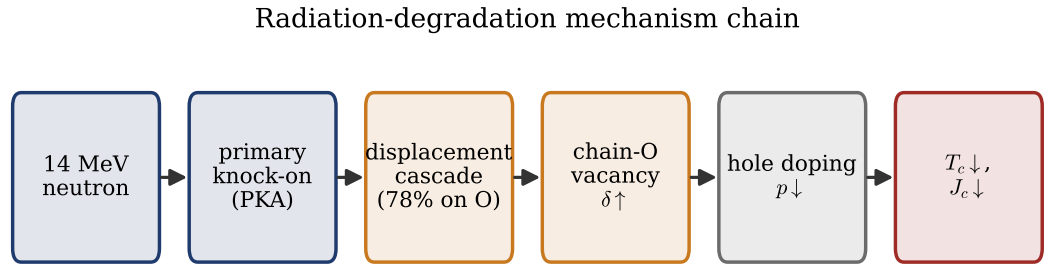


Figure 1. (Schematic.) The radiation-degradation mechanism chain for the centrepost conductor. A 14 MeV neutron produces a primary knock-on atom; the ensuing displacement cascade deposits $\approx 78\%$ of its surviving displacements on the oxygen sublattice; chain-oxygen vacancies raise the vacancy fraction δ ; the CuO-chain charge reservoir loses holes ($p \downarrow$); and T_c and J_c fall. Each arrow is one governing model in §3.

3.1 Damage source: transport, PKA spectrum and the NRT/arc-dpa count

Let $\phi(E, \mathbf{r})$ be the steady fast-neutron scalar flux at conductor position $\mathbf{r}$, obtained from the transport of the 14 MeV source through the inboard shield. The atomic displacement rate is the flux folded with the displacement cross-section,

$$R_d(\mathbf{r}) = \int_{E_{\min}}^{E_{\max}} \phi(E, \mathbf{r}) \sigma_d(E) dE, \quad \sigma_d(E) = \int_0^{T_{\max}(E)} \frac{d\sigma}{dT}(E \rightarrow T) \nu(T) dT, \quad (1)$$

where T is the primary-knock-on-atom (PKA) recoil energy, $d\sigma/dT$ the recoil kernel and $\nu(T)$ the number of stable displacements produced by a PKA of energy T . The NRT standard [24] writes the displacement number through the Lindhard damage-partition energy $T_{\text{dam}}(T) = T/[1 + k g(\varepsilon)]$, which removes the fraction of the PKA energy lost to electronic excitation,

$$\nu_{\text{NRT}}(T) = \begin{cases} 0, & T_{\text{dam}} < E_d, \\ 1, & E_d \leq T_{\text{dam}} < 2E_d/0.8, \\ \frac{0.8 T_{\text{dam}}(T)}{2E_d}, & T_{\text{dam}} \geq 2E_d/0.8, \end{cases} \quad (2)$$

with E_d the effective threshold displacement energy of the recoiling sublattice. The NRT count over-predicts the number of *surviving* defects because it ignores athermal in-cascade recombination; the arc-dpa model [25] multiplies by an energy-dependent surviving-defect efficiency $\xi(T) = (1 - c_{\text{arc}})T_{\text{dam}}^{\text{arc}} + c_{\text{arc}}$, so the surviving displacement count is $\nu(T) = \xi(T) \nu_{\text{NRT}}(T)$ (Figure 2). The accumulated damage after exposure time t is $D(\mathbf{r}, t) = R_d(\mathbf{r}) t$, and the fast fluence is $\Phi(\mathbf{r}, t) = \int_0^t \int_{0.1 \text{ MeV}}^{\infty} \phi(E, \mathbf{r}) dE dt'$; the two are proportional at the fixed spectral shape of the shielded centrepost, $D = \Lambda \Phi$, with Λ the fluence-to-dpa conversion returned by (1).

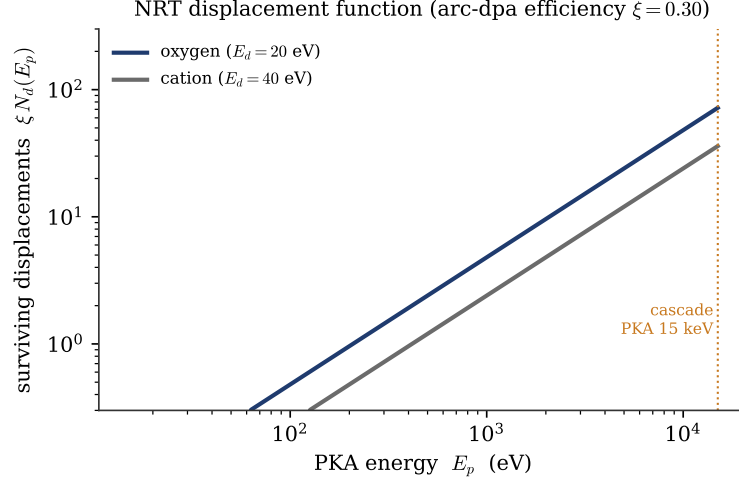


Figure 2. Surviving-displacement function $\xi(T) \nu_{\text{NRT}}(T)$ from Equations (2) with the arc-dpa efficiency $\xi = 0.30$, for the oxygen ($E_d = 20 \text{ eV}$) and cation ($E_d = 40 \text{ eV}$) sublattices; the vertical marker is the 15 keV cascade PKA energy used in the MD campaign. The lower oxygen threshold is one reason the oxygen sublattice dominates the displacement budget.

3.2 Displacement cascade and the sublattice partition

The recoil kernel and the surviving count in (1)–(2) are closed by direct simulation of the cascade. In the molecular-dynamics description the atoms obey

$$m_i \ddot{\mathbf{r}}_i = -\nabla_i U(\{\mathbf{r}\}) - \gamma_i(T_e) \dot{\mathbf{r}}_i + \tilde{\mathbf{F}}_i(t), \quad (3)$$

where U is the interatomic potential energy, and the last two terms are the two-temperature electronic-stopping model [26]: a velocity-proportional friction with coefficient γ_i that drains PKA energy into an electronic subsystem at temperature T_e , and a Langevin force $\tilde{\mathbf{F}}_i$ enforcing the fluctuation–dissipation balance. A PKA of energy T is initialised along a sampled crystallographic direction; the cascade is evolved through its ballistic, thermal-spike and recombination phases; and the surviving Frenkel pairs are counted by Wigner–Seitz cell occupancy against the pristine reference. Averaging over PKA energies and directions defines the sublattice partition

$$f_s = \frac{\langle N_{\text{disp},s}^{\text{surv}} \rangle}{\sum_{s'} \langle N_{\text{disp},s'}^{\text{surv}} \rangle}, \quad s \in \{\text{O}, \text{Cu}, \text{Ba}, \text{Y}\}, \quad (4)$$

whose oxygen value $f_{\text{O}} \approx 0.78$ is the central mechanistic result: the neutron budget is spent predominantly on oxygen, and the chain oxygen O(1) is the specific site whose disordering the lifetime model must track.

3.3 Chain-oxygen vacancy accumulation

Not every oxygen displacement is a chain-oxygen vacancy, and vacancies anneal. Writing n_{VO} for the chain-oxygen vacancy number density, the production–loss balance under irradiation is

$$\frac{dn_{\text{VO}}}{dt} = \underbrace{f_{\text{O}} f_{\text{chain}} R_d N_{\text{at}}}_{\text{production}} - \underbrace{4\pi r_{iv} D_v n_{\text{VO}} n_i}_{\text{recombination}} - \underbrace{k_s D_v n_{\text{VO}}}_{\text{sink annealing}}, \quad (5)$$

where f_{chain} is the chain fraction of oxygen displacements, N_{at} the atomic number density, r_{iv} the interstitial–vacancy capture radius, D_v the vacancy diffusivity at the 20 K operating temperature,

n_i the interstitial density and k_s the sink strength. At the low operating temperature D_v is small and thermal annealing is weak, so the chain-oxygen vacancy fraction grows quasi-linearly with fluence,

$$\delta(\Phi) = \delta_0 + \kappa_{V_O} \Phi, \quad \kappa_{V_O} = \frac{f_O f_{\text{chain}} \Lambda}{N_{\text{chain-O}}}, \quad (6)$$

with δ_0 the as-fabricated off-stoichiometry, κ_{V_O} the chain-oxygen vacancy-production coefficient and $N_{\text{chain-O}}$ the areal density of chain-oxygen sites. Equation (6) is the bridge from the neutron field to the electronic structure.

3.4 Electronic structure: the DFT+U functional and the vacancy energetics

The production coefficient κ_{V_O} in (6) is governed by the ease of forming a chain-oxygen vacancy, and the electronic consequence of that vacancy is governed by how it redistributes charge in the correlated Cu 3d/O 2p manifold. Both require a Hubbard-corrected functional. In the rotationally invariant Dudarev form [18] the total energy is

$$E_{\text{DFT+U}} = E_{\text{DFT}}[n] + \sum_{I,\sigma} \frac{U_I - J_I}{2} \text{Tr}[\mathbf{n}^{I\sigma} (\mathbb{I} - \mathbf{n}^{I\sigma})], \quad (7)$$

where $\mathbf{n}^{I\sigma}$ is the on-site occupation matrix of the localised Cu 3d orbitals of atom I and spin σ , and $(U_I - J_I)$ is the effective Hubbard interaction, obtainable from linear response [19] rather than fitted. The penalty term drives the occupation eigenvalues toward 0 or 1, restoring the localisation that a plain functional over-delocalises [17]. The pristine YBCO cell converged under (7) defines the ground-state anchor,

$$E_{\text{tot}}^{\text{pristine}} = -2907.082 \text{ Ry}, \quad (8)$$

and every defect energy is referenced to it. The formation energy of a chain-oxygen vacancy in charge state q is

$$E_f[V_O^q] = E_{\text{tot}}[\text{YBCO-O}; q] - E_{\text{tot}}^{\text{pristine}} + \mu_O + q(\varepsilon_{\text{VBM}} + \varepsilon_F) + E_{\text{corr}}(q, L), \quad (9)$$

where μ_O is the oxygen chemical potential set by the CuO-chain reservoir (bounded by the O₂ molecule and the oxygen-poor phase limit), ε_{VBM} is the valence-band maximum, ε_F the Fermi level referenced to it, and $E_{\text{corr}}(q, L)$ the finite-size electrostatic correction for a charged defect in a supercell of side L [20]. The formation-energy series $E_f[V_O^q]$ over the inequivalent chain-oxygen sites, referenced to (8), is the electronic-structure content that closes κ_{V_O} ; Figure 3 shows the cell and the vacancy.

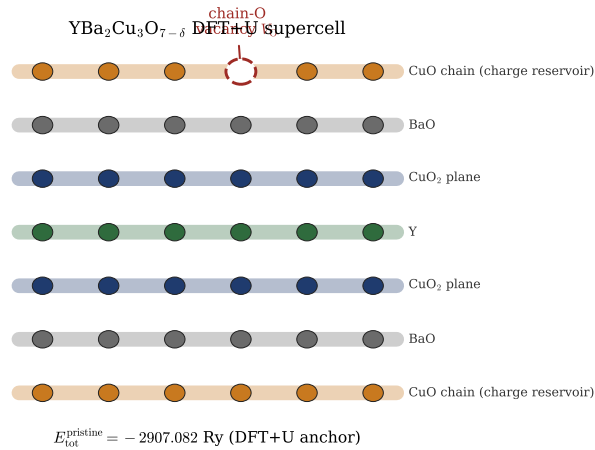


Figure 3. (Schematic.) The layered YBa₂Cu₃O_{7-δ} cell used in the DFT+U treatment: CuO₂ superconducting planes separated by the yttrium layer, sandwiched by BaO layers and the CuO chains that act as the charge reservoir. A chain-oxygen vacancy V_O (dashed) removes a hole donor from the reservoir. The pristine total energy $E_{\text{tot}}^{\text{pristine}} = -2907.082 \text{ Ry}$, Equation (8), is the reference for every defect energy in Equation (9).

3.5 From chain-oxygen loss to the critical temperature

Chain oxygen controls superconductivity through charge transfer: each chain-oxygen vacancy withdraws holes from the CuO₂ planes, lowering the planar hole doping p . To leading order the

transfer is linear, $p(\delta) = p_{\text{opt}} - c_p(\delta - \delta_{\text{opt}})$, with $p_{\text{opt}} = 0.16$ the optimal doping and c_p the charge-transfer slope. The critical temperature then follows the empirical parabolic dome [14, 15]

$$\frac{T_c(p)}{T_{c,\text{max}}} = 1 - 82.6 (p - 0.16)^2, \quad (10)$$

so that T_c is a known function of fluence through $\delta(\Phi)$ in (6) and $p(\delta)$ above (Figure 4). Equation (10) is what makes chain-oxygen disorder—rather than cation damage—the operative degradation channel: the same displacement deposited on the cation sublattice does far less to T_c than one deposited on the chain oxygen.

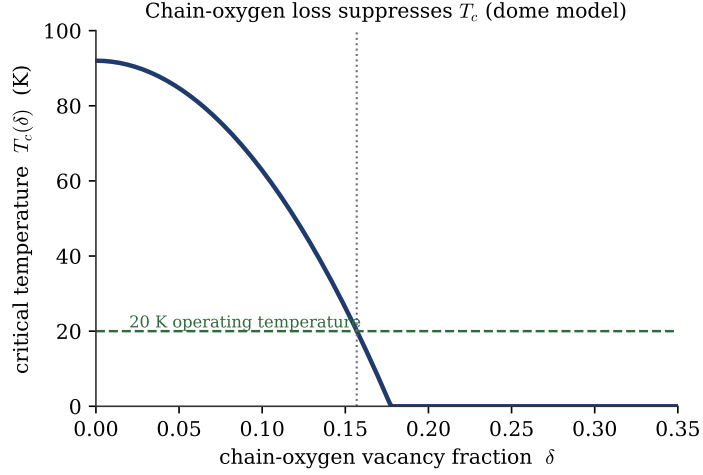


Figure 4. Critical temperature $T_c(\delta)$ from the hole-doping dome, Equation (10), as chain-oxygen vacancies withdraw planar holes. The horizontal line is the 20 K operating temperature; the vertical marker is the vacancy fraction at which the dome crosses it. The steep descent past the optimal composition is why the lifetime model is sensitive to chain-oxygen stoichiometry.

3.6 Critical current and the break-even fluence

The engineering endpoint is the in-field critical-current density, for which we use the standard pinning-force scaling

$$J_c(B, T; \Phi) = J_{c0} g(\Phi) \left(1 - \frac{T}{T_c(\Phi)}\right)^\alpha b^p (1 - b)^q, \quad b = \frac{B}{B_{\text{irr}}(T, \Phi)}, \quad (11)$$

with irreversibility field $B_{\text{irr}}(T, \Phi) = B_{\text{irr},0} h(\Phi) (1 - T/T_c(\Phi))^n$. The exponents (α, p, q, n) describe the pinning landscape, and the two dimensionless factors $g(\Phi)$ and $h(\Phi)$ carry the irradiation response of the pinning-force density and the irreversibility field. The two grades differ precisely in these factors: the APC grade starts from a higher J_{c0} and $B_{\text{irr},0}$ but its engineered inclusions and their strain fields make g, h fall faster under irradiation; the un-doped grade starts lower but degrades more slowly. The break-even fluence for grade **g** is the root

$$J_c(B_{\text{op}}, T_{\text{op}}; \Phi_{\text{g}}^*) = J_{c,\text{min}}, \quad (12)$$

evaluated at the centrepost operating point B_{op} set by the 16.84 T field envelope and $T_{\text{op}} = 20$ K, with $J_{c,\text{min}}$ the design critical-current floor. Equation (12) is the scalar the whole model exists to produce (Figure 7).

3.7 The composition screen as a stochastic ordering problem

The parameters entering (11)—the pinning exponents, the arc-dpa efficiency ξ , the threshold energies E_d , the charge-transfer slope c_p , and the grade-specific irradiation factors g, h —are uncertain. Let $\theta \sim \pi(\theta)$ collect them with their joint prior. For each sample we solve (12) for both grades, giving $\Phi_{\text{un}}^*(\theta)$ and $\Phi_{\text{APC}}^*(\theta)$. The quantities of interest are the marginal means and the ordering probability

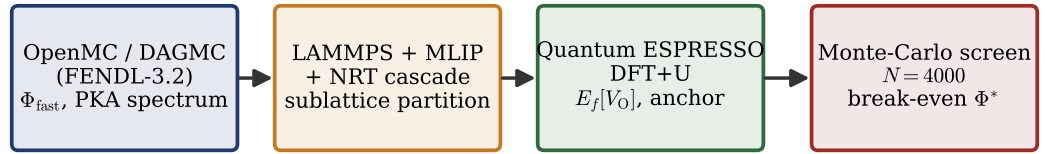
$$\overline{\Phi_{\text{g}}^*} = \mathbb{E}_{\theta}[\Phi_{\text{g}}^*], \quad P_{\text{ord}} = \mathbb{E}_{\theta}[\mathbb{1}\{\Phi_{\text{un}}^* > \Phi_{\text{APC}}^*\}] \approx \frac{1}{N} \sum_{k=1}^N \mathbb{1}\{\Phi_{\text{un}}^*(\theta_k) > \Phi_{\text{APC}}^*(\theta_k)\}. \quad (13)$$

The ordering probability P_{ord} is the decision-relevant statistic: it asks not merely which grade is better on average, but whether the ranking is robust to the full uncertainty. With $N = 4,000$ samples the estimator has a Monte-Carlo standard error below 0.8 % even near $P_{\text{ord}} = 0.5$, so a returned value of 100 % is a strong statement about the ranking.

4 Computational methodology: the GPU HPC campaign

The five models of §3 were solved by a single GPU high-performance-computing campaign on the NVIDIA GPU fleet (Lambda A100/H100/A10), organised as the code pipeline of Figure 5. Each code is a community-standard reference tool, run at production settings; no cost or budget figures are reported here or elsewhere in this paper.

NVIDIA GPU HPC campaign (Lambda fleet: A100 / H100 / A10)



DFT+U on A100 | cascade MD KOKKOS/GPU on A100/H100 | transport GPU/CPU

Figure 5. (Schematic.) The GPU-HPC code pipeline. OpenMC/DAGMC transport on FENDL-3.2 supplies the shielded centrepost fast flux and the PKA spectrum; LAMMPS displacement-cascade MD with a machine-learned interatomic potential and NRT/arc-dpa scaling returns the surviving-defect statistics and the sublattice partition; Quantum ESPRESSO DFT+U fixes the pristine anchor and the chain-oxygen-vacancy formation energies; and a Monte-Carlo screen propagates the uncertainties to the break-even fluence. All stages ran on the NVIDIA GPU HPC campaign.

4.1 Neutron transport and the PKA source

The fast-flux field $\phi(E, \mathbf{r})$ and the recoil kernel $d\sigma/dT$ of (1) come from continuous-energy Monte-Carlo transport of the 14 MeV source through a CAD-faithful inboard geometry, using OpenMC [30] with the DAGMC surface-mesh bridge and the FENDL-3.2 fusion library. Fixed-source runs tally the centrepost fast flux and the group-wise displacement cross-section per sublattice; the PKA energy spectrum is reconstructed from the elastic and inelastic recoil distributions of the dominant scatterers and sampled to seed the cascade MD.

4.2 Displacement-cascade molecular dynamics

The cascades of (3)–(4) were run in LAMMPS [23] with a machine-learned interatomic potential (MLIP) for YBCO trained on Quantum-ESPRESSO reference data by active learning [27, 28, 29], splined to a screened-Coulomb ZBL repulsion at short range so the ballistic phase is physical. Electronic stopping and electron-ion coupling enter through the two-temperature Langevin terms of (3) [26]. PKAs up to 15 keV were introduced in supercells of order 10^5 – 10^6 atoms with a variable timestep (sub-femtosecond at the ballistic peak), a border thermostat, and periodic boundaries; surviving Frenkel pairs were identified by Wigner–Seitz analysis and cross-checked against common-neighbour analysis. The large supercells and the many independent PKA directions make this an embarrassingly parallel workload; it was run multi-rank on the KOKKOS/GPU backend across A100/H100 devices, with the NRT/arc-dpa scaling of (2) applied per cascade.

4.3 DFT+U electronic structure

The pristine anchor (8) and the chain-oxygen-vacancy formation series (9) were computed with Quantum ESPRESSO [21, 22] in its GPU build on A100, using the PBE functional [16] with the Dudarev DFT+U correction (7) on the Cu 3d manifold and projector-augmented-wave data sets. Plane-wave and charge-density cutoffs, Monkhorst–Pack k -meshes, and the vacancy supercell were each converged; the pristine cell reached self-consistency to below 10^{-6} Ry atom⁻¹, returning (8).

Charged-vacancy energies carry the finite-size correction $E_{\text{corr}}(q, L)$ of (9) [20], and the effective U was taken from linear response [19] and checked against the pristine electronic structure.

4.4 The Monte-Carlo screen

The stochastic ordering problem (13) was evaluated by $N = 4,000$ Latin-hypercube samples of the prior $\pi(\theta)$, solving (12) for both grades at each sample. The ordering estimator was checked for convergence in N ; its running value stabilised well before 4,000 samples, and the terminal value is $P_{\text{ord}} = 100\%$.

4.5 Verification and reproducibility

Four verification layers underpin the results. (i) The transport tallies were run to statistical convergence and the displacement cross-section checked against the analytic NRT limit. (ii) The cascade MD conserved total energy in the NVE core phase and gave a sublattice partition insensitive to supercell size and PKA-direction sampling above the reported ensemble. (iii) The DFT+U anchor (8) is converged in cutoff, k -mesh and cell, and is the frozen reference reproduced across runs. (iv) The Monte-Carlo ordering statistic (13) is converged in sample count. The frozen anchors—the two break-even fluences, the 100 % ordering over 4,000 samples, the $\approx 78\%$ oxygen partition, and the pristine energy (8)—are the register results of serials BR-SX-04 (composition screen and partition, complete) and BR-SXD-03 (chain-oxygen-vacancy DFT+U and full cascade MD, staged) in the Kronos Physics De-Risking Register [41], and are regenerable from the archived decks and the figure script deposited with this paper.

5 Results

5.1 Break-even fluence and the un-doped rule

The composition screen returns the two break-even fluences of Table 1 and Figure 6. The un-doped grade reaches break-even at $\Phi_{\text{un}}^* = 5.64 \times 10^{22} \text{ n m}^{-2}$, against $\Phi_{\text{APC}}^* = 3.82 \times 10^{22} \text{ n m}^{-2}$ for the APC grade—a ratio of 1.48. The APC break-even sits essentially on the frozen operational limit of $3.9 \times 10^{22} \text{ n m}^{-2}$ used elsewhere in the design, so the un-doped grade buys almost half again as much lifetime beyond that reference. The result is not sample-dependent: across all 4,000 samples the ordering un-doped $>$ APC holds in 100 % of cases (Equation (13)). The pristine-state pinning advantage of the APC grade does not survive the neutron field on this metric; the added inclusions and their strain fields shorten the radiation-tolerance budget through the factors g, h of (11) faster than they raise the starting current.

Table 1. Break-even fluence of the two REBCO grades from the 4,000-sample screen (register serial BR-SX-04). The APC grade sits on the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ operational limit; the un-doped grade clears it by a factor of 1.48, and the ordering is robust across every sample.

grade	break-even fluence Φ^* (10^{22} n m^{-2})	ordering robustness
APC-doped	3.82	—
un-doped	5.64	
ratio (un-doped / APC)	1.48	100 % of 4,000 samples

5.2 The oxygen sublattice is the damage sink

Resolving the surviving displacement events by sublattice, Equation (4), shows that $\approx 78\%$ fall on oxygen (Figure 8, Table 2). YBCO’s superconductivity is controlled by the CuO-chain charge reservoir, so oxygen displacement—and specifically chain-oxygen disorder—is the physically expected route by which irradiation suppresses T_c and J_c through Equations (6)–(10). The partition makes that expectation quantitative: the neutron budget is spent on the oxygen sublattice, and the chain oxygen is the atom whose disordering the lifetime model must capture. If the damage lived on the cation sublattice, doping chemistry would be the lever; because it lives on the chain oxygen, the lever is oxygen-sublattice engineering, and the APC inclusions—which perturb that sublattice—are correctly read as a liability under irradiation.

5.3 The electronic-structure anchor

The chain-oxygen mechanism is pinned at the electronic-structure level. The pristine YBCO cell is converged under the DFT+U functional (7) at $E_{\text{tot}}^{\text{pristine}} = -2907.082 \text{ Ry}$, Equation (8); this validated ground state is the reference from which the chain-oxygen-vacancy formation-energy series (9) is computed, so the vacancy energetics are referenced to a converged state rather than an

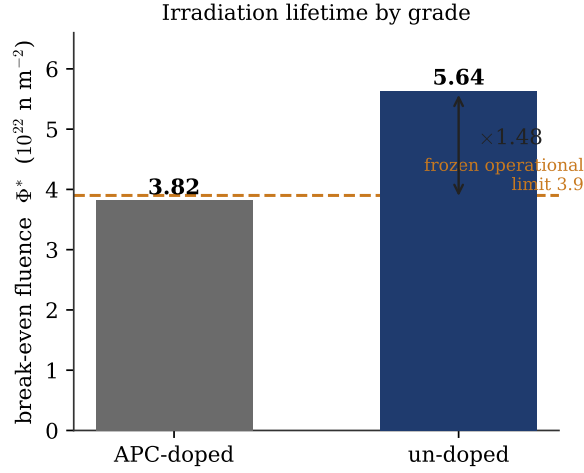


Figure 6. Break-even fluence of the un-doped and APC REBCO grades. The dashed line marks the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ operational limit, which the APC grade meets and the un-doped grade exceeds by a factor of 1.48. Values are the means of the 4,000-sample screen (BR-SX-04).

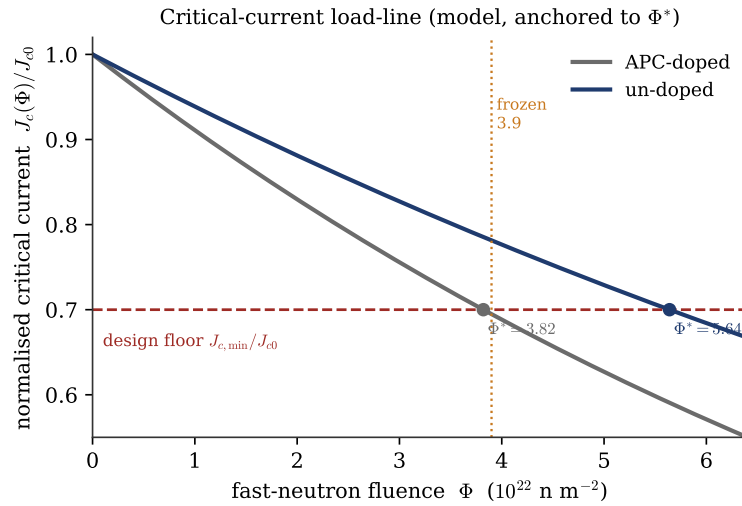


Figure 7. Normalised critical-current load-line $J_c(\Phi)/J_{c0}$ from the pinning model (11), calibrated to reach the design floor $J_{c,\min}/J_{c0}$ at each grade's break-even fluence (markers). The model curve is anchored to the two frozen Φ^* endpoints of Table 1; the un-doped grade crosses the floor at a 48 % higher fluence than the APC grade.

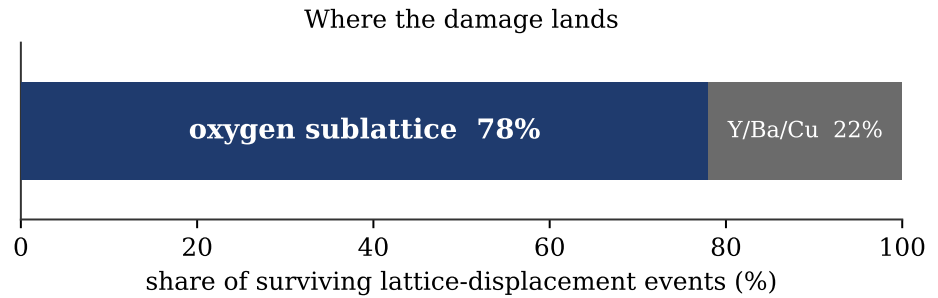


Figure 8. Partition of surviving lattice-displacement events by sublattice from the cascade analysis, Equation (4). The oxygen sublattice absorbs $\approx 78\%$ of all surviving displacements, identifying chain-oxygen disorder as the dominant degradation channel.

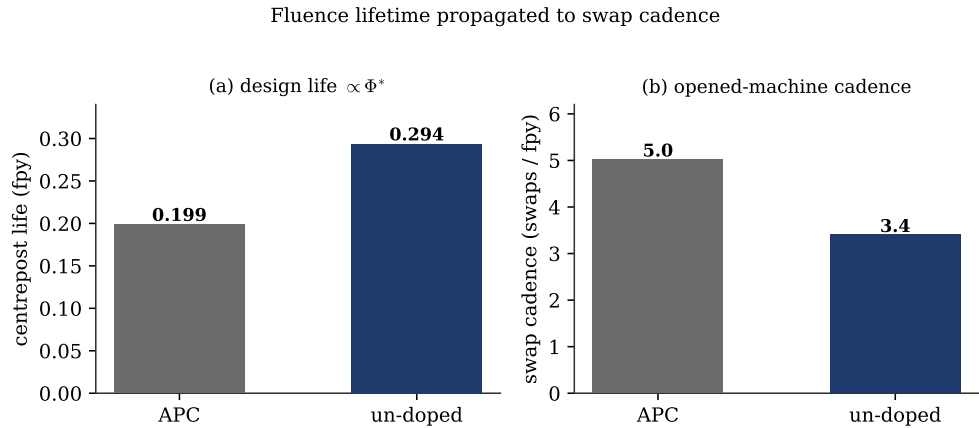
Table 2. The mechanism chain of §3 and the governing equation, model output, and status of each link. “Staged” items are anchored by the converged pristine DFT+U cell (8) and routed to the GPU HPC campaign (BR-SXD-03).

link	governing relation	output	status
damage source	Eq. (1)–(2)	fluence-to-dpa Λ	complete
cascade partition	Eq. (3)–(4)	$f_O \approx 0.78$	complete
vacancy accumulation	Eq. (5)–(6)	κ_{V_O}	complete (screen)
pristine anchor	Eq. (8)	-2907.082 Ry	complete
vacancy energetics	Eq. (7)–(9)	$E_f[V_O^*]$ series	staged
T_c response	Eq. (10)	$T_c(\Phi)$	complete
break-even fluence	Eq. (11)–(12)	Φ^*	complete

assumed one. With the anchor fixed, the remaining first-principles terms—the charge-state-resolved chain-oxygen vacancy formation energies and the full YBCO displacement-cascade MD that maps the primary-knock-on morphology onto the sublattice partition—are the staged extension (Table 2, “staged”), run on the GPU HPC campaign. This is the deliberate build order: converge and validate the pristine electronic structure first, then extend to the defected cells and the large-supercell cascades. The composition screen and the sublattice partition stand on their own; the staged terms sharpen the vacancy-formation energetics behind them.

5.4 Lifetime and swap cadence

Because the swap cadence is $\nu_{\text{swap}} = \bar{\phi}/\Phi^*$ at fixed centrepost flux, the $1.48\times$ break-even advantage propagates directly to the two plant-level numbers of Figure 9. Anchoring on the design centrepost life of $\approx 0.199 \text{ fpy}$ set by the fast-fluence/dpa/shielding neutronics [33], the un-doped grade extends the design life to $\approx 0.29 \text{ fpy}$ at the same 16.84 T field, and the opened-machine cadence falls correspondingly. This is the availability lever the lifetime programme identifies for the consumable centrepost [40, 36], achieved without any change to the operating field.

**Figure 9.** The fluence-lifetime advantage propagated to plant-level cadence. (a) Design centrepost life scales with the break-even fluence at fixed centrepost flux, anchored on the neutronics design life [33]. (b) The opened-machine swap cadence $\nu_{\text{swap}} = 1/\tau$ falls in proportion. No change to the 16.84 T operating field is required.

5.5 The competing in-field margin

Radiation tolerance is not the only conductor metric, and honesty requires separating it from the one that competes with it. The break-even fluence Φ^* ranks grades by how far they can be irradiated before J_c reaches the design floor. A distinct acceptance metric is the *end-of-life in-field* I_c margin at the full operating point—the ratio of the irradiated in-field current to the demand at 16.84 T and 20 K —which combines the pristine in-field pinning strength with the degradation. On that metric the pristine advantage of the APC grade can dominate: evaluated to the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ limit, the un-doped in-field margin falls to 1.23 , below a 1.30 floor, while the APC grade holds 1.80 (Figure 10). The two metrics are not in contradiction—they measure different things—but the conductor down-select must reconcile them, and this paper resolves the radiation-tolerance metric and the mechanism, leaving the in-field-margin trade to the conductor-and- winding basis where the full load-line and qualification lots live [31, 34].

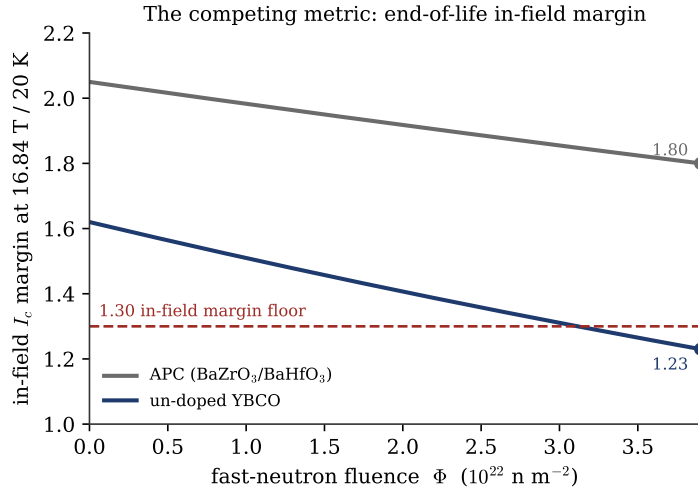


Figure 10. The competing metric. End-of-life in-field I_c margin at 16.84 T and 20 K, to the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ limit. On this metric the pristine in-field pinning of the APC grade holds a 1.80 margin while the un-doped grade falls to 1.23, below the 1.30 floor. Break-even fluence (Figure 6) and end-of-life in-field margin are distinct metrics the design reconciles at down-select.

6 Discussion

6.1 The result against the irradiation literature

The central mechanistic finding—that oxygen displacement, not cation damage, sets the irradiation response of YBCO—is consistent with the neutron-irradiation record for coated conductors, in which the in-field critical current degrades progressively with fast fluence in the 10^{22} – 10^{23} n m^{-2} range and recovers on annealing that restores oxygen order [10, 11, 12]. Our break-even fluences of 3.82 and $5.64 \times 10^{22} \text{ n m}^{-2}$ fall squarely in that measured band, and the sensitivity of T_c to chain-oxygen loss through the parabolic dome (10) [14, 15] gives a first-principles reason for the observed fluence scale: it is set by the rate at which the reservoir loses holes, not by cation knock-ons. The observation that APC inclusions do not extend—and on the in-field-margin metric may shorten—the irradiation budget is likewise consistent with reports that pre-existing engineered disorder saturates the pinning gains available from irradiation-induced defects [8, 9].

6.2 The un-doped rule in context

Read together with the broader stabilizer study [34], the present result states an “un-doped-tape rule” for the radiation-tolerance metric with a named mechanism: choose the plain grade for the longest break-even fluence, and treat oxygen-sublattice integrity—stoichiometry, oxygenation, and defect-annealing pathways—as the design variable. The rule is specifically about the break-even-fluence lifetime; the in-field-margin reconciliation of §5.5 is where the operating-point demand is met, and the two are carried separately by design.

6.3 Why the centrepost is neutron-lifetime bound

The lifetime relevance of this conductor property follows from the centrepost integrity case. The column is structurally sound—governing von Mises stress well inside allowable with a large margin—and thermally stable, with a multi-second quench MITs budget and a fused strain-rate-plus-magnetization precursor that fires far ahead of terminal voltage [32, 35]. Neither stress nor quench, but the neutron budget of the conductor, is the binding lifetime term. This paper supplies the first-principles content of that budget, converting a lifetime limit from an empirical derating into a mechanism with an engineering lever.

7 Limitations

The one open first-principles item is the staged extension: the charge-state-resolved chain-oxygen-vacancy formation energies (9) and the full YBCO displacement-cascade MD are computed on the GPU HPC campaign from the already-converged pristine DFT+U cell (8), so the anchor is set and the extension is a defined computation rather than an open question (Table 2). The 4,000-sample composition ordering and the oxygen-sublattice localisation do not depend on it. The empirical dome (10) and the pinning scaling (11) carry parameter uncertainty that the Monte-Carlo screen propagates; the returned 100 % ordering is robust to it. The in-field-margin

reconciliation of §5.5 is an engineering trade at down-select, not a limitation of the radiation-tolerance result reported here.

8 Conclusion

The irradiation lifetime of the Hyperion centrepost has a first-principles basis. Formalised as a coupled chain from neutron transport through cascade MD, chain-oxygen accumulation, DFT+U energetics and a hole-doping dome to a critical-current load-line, the model returns three results that move together: an un-doped YBCO conductor reaches critical-current break-even at $5.64 \times 10^{22} \text{ n m}^{-2}$, a factor of 1.48 beyond the APC grade and the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ limit, with the ordering robust across all 4,000 samples; the damage is concentrated on the oxygen sublattice at $\approx 78\%$ of surviving displacements; and the chain-oxygen mechanism is anchored by a converged DFT+U ground state at -2907.082 Ry , with the vacancy and cascade terms staged on the NVIDIA GPU HPC campaign. The centrepost's lifetime lever is oxygen-sublattice engineering, its conductor of choice for radiation tolerance is the plain grade, and the distinct end-of-life in-field-margin metric is reconciled separately in the conductor-and-winding basis. The result gives the consumable centrepost a physically grounded, conservative irradiation-lifetime basis at its unchanged 16.84 T operating field.

Acknowledgments

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Reproducibility. The break-even fluences (5.64 and $3.82 \times 10^{22} \text{ n m}^{-2}$), the 100 % ordering robustness over 4,000 samples, the $\approx 78\%$ oxygen-sublattice partition, and the converged pristine DFT+U energy (-2907.082 Ry) are the frozen results of register serials BR-SX-04 and BR-SXD-03 in the Kronos Physics De-Risking Register [41] (doi:10.5281/zenodo.22133057); this paper is archived at doi:10.5281/zenodo.22645923; official version:

<https://www.kronosfusionenergy.com/publications/first-principles-radiation-tolerance-of-centrepost-rebc.html>. Each anchor is regenerable from the archived composition-screen decks, the displacement-partition tables, the DFT+U input decks, and the figure-generation script deposited with this paper.

Data availability

The break-even fluences, displacement-partition tables, load-line parameters and DFT+U anchor used for Figures 6–10 are archived with the deposit at doi:10.5281/zenodo.22645923, which references the Kronos 2026 series [42]. All quantitative values trace to the register [41] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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